



VISVESVARAYA TECHNOLOGICAL UNIVERSITY

SKILL DEVELOPMENT ACTIVITY

22MCA335: NOSQL

On

Blood Donor Management System

Submitted by

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Under the guidance of

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CERTIFICATE

This is to certify that **Keerthana B V** has undertaken this Skill Development Activity entitled “**Blood Donor Management System**” for the partial fulfillment of the completion of SDA assessment. Student of 3rd semester, in partial fulfillment for the Skill Development Activity–NOSQL (22MCA335), during the academic year 2024-2025.

Faculty In-charge

TAMAL SARKAR

DECLARATION

I hereby declare that the project report entitled **Blood Donor Management System** based on study undertaken by me, towards the partial fulfillment for the Skill Development Activity – NOSQL (22MCA335), carried out during the 3rd semester, has been compiled purely from the academic point of view and is therefore, presented in a true and sincere academic spirit. Contents of this report are based on my original study and findings in relation there to are neither copied nor manipulated from other reports or similar documents, either in part or in full, and it has not been submitted earlier to any University/College/Academic institution for the award of any Degree and that the work has not been published in any specific or popular magazines.

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1.ABSTRACT

The Blood Donor Management System is a comprehensive web application designed to streamline the process of blood donation management in healthcare facilities. This system addresses the critical need for efficient blood resource management by connecting blood donors with recipients in need. The application leverages MongoDB, a NoSQL database, to provide flexibility in data storage and retrieval, making it particularly suitable for healthcare applications where data structures may evolve over time.

The system features donor and recipient management, blood type matching algorithms, donation tracking, and a user-friendly interface for administrators. By implementing this system, healthcare facilities can significantly reduce the time required to find compatible blood donors, especially in emergency situations, potentially saving lives through faster response times.

This details the development, implementation, and evaluation of the Blood Donor Management System, highlighting its key features, technical architecture, and the specific advantages of using MongoDB as the database solution. The system demonstrates how modern web technologies and NoSQL databases can be effectively utilized to solve real-world healthcare challenges.

2. INTRODUCTION

Blood donation is a critical component of healthcare systems worldwide, with millions of patients requiring blood transfusions annually for surgeries, treatments, and emergency care. Despite the importance of blood donation, many healthcare facilities still rely on outdated, paper-based systems or disconnected databases to manage their blood donation processes. This leads to inefficiencies, delays in finding compatible donors, and potential errors in matching donors with recipients.

The Blood Donor Management System addresses these challenges by providing a centralized, digital platform for managing the entire blood donation process.

The system is designed with :

1. **Healthcare Administrators** : Who need to efficiently manage blood resources and track donation history
2. **Medical Staff** : Who require quick access to compatible blood donors in emergency situations
3. **Blood Donors** : Whose information needs to be accurately recorded and maintained
4. **Recipients** : Whose blood type and urgency level must be matched with appropriate donors

The system is built using modern web technologies, with a React-based frontend for a responsive user interface and a Python Flask backend API that interacts with a MongoDB database. This technology stack was chosen specifically to provide flexibility, scalability, and ease of development.

MongoDB, as a NoSQL database, offers several advantages for this application:

1. **Schema Flexibility** : Blood donor and recipient data may contain varying fields and attributes
2. **Document-Oriented Structure** : Naturally maps to the hierarchical relationships in healthcare data
3. **Scalability** : Can handle growing volumes of donor and recipient records
4. **Query Performance** : Provides efficient querying for complex matching operations

3. OBJECTIVES

The Blood Donor Management System was developed with the following key objectives:

- 1. Streamline Donor Management :** Create a centralized system to register, track, and manage blood donors, including their personal information, blood type, donation history, and availability status.
- 2. Efficient Recipient Management:** Develop functionality to register and manage blood recipients, including their blood type requirements, urgency levels, and contact information.
- 3. Automated Matching System:** Implement an intelligent matching algorithm that can quickly identify compatible donors for recipients based on blood type compatibility, location, and availability.
- 4. Donation Tracking:** Create a comprehensive tracking system for donations from initiation to completion, including status updates and historical records.
- 5. User-Friendly Interface:** Design an intuitive administrative interface that allows healthcare staff to efficiently manage the entire blood donation process.
- 6. Data Security and Privacy:** Ensure that sensitive donor and recipient information is securely stored and accessed only by authorized personnel.

Why NoSQL (MongoDB) is Required for this Problem

The Blood Donor Management System specifically requires a NoSQL database like MongoDB for several critical reasons:

1. Flexible Schema Design: Unlike traditional blood management systems that use rigid SQL databases, our system needs to accommodate varying data attributes for donors and recipients. For example:

1. Some donors may provide additional health information while others may not
2. Recipients may have different urgency levels and specific requirements
3. MongoDB's schema-less design allows us to store these varying attributes without requiring database schema modifications

2. Document-Oriented Structure : The hierarchical nature of blood donation data (donors with donation histories, recipients with medical requirements) maps naturally to MongoDB's document model, allowing for:

1. Embedding donation history directly within donor records
2. Storing complex recipient requirements as nested documents
3. Maintaining relationships between donors, recipients, and donations without complex joins

3. Query Performance for Matching Operations : Blood type matching requires complex queries that can be efficiently executed in MongoDB:

1. Finding donors with compatible blood types (e.g., O- donors for multiple recipient types)
2. Filtering by location, availability, and last donation date
3. MongoDB's indexing capabilities optimize these complex queries

4. Scalability for Growing Healthcare Data : As the system accumulates more donor and recipient records over time:

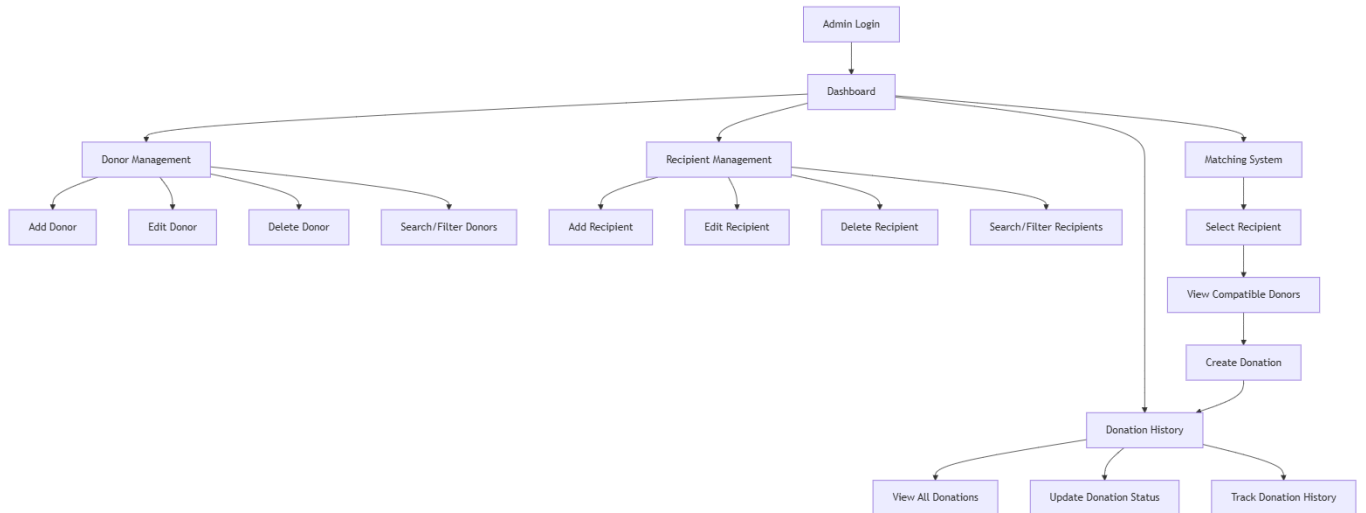
1. MongoDB's horizontal scaling capabilities allow the system to grow
2. Sharding can distribute data across multiple servers if needed
3. This ensures the system remains responsive even with large datasets

These specific requirements make a NoSQL database like MongoDB significantly more suitable for this application than a traditional relational database, which would require complex table structures, joins, and frequent schema modifications to accommodate the same functionality.

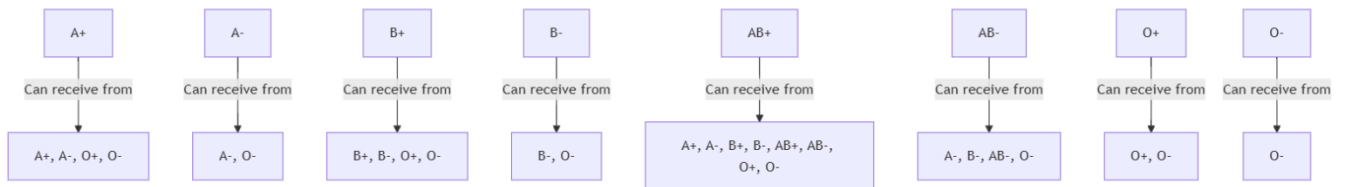
4. METHODOLOGY

a. Flow Diagram

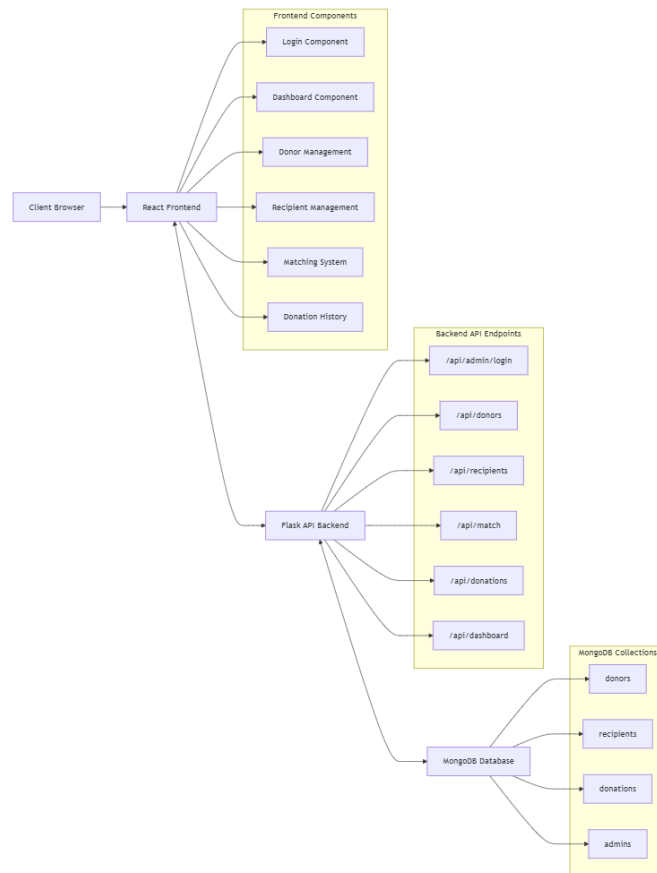
Blood Donor Management System Flow



Blood Type Compatibility Chart



Blood Type Compatibility Chart



b. Framework

The Blood Donor Management System is built using a modern technology stack that ensures flexibility, scalability, and ease of development:

Frontend Framework:

- React.js: A JavaScript library for building user interfaces
- Bootstrap: For responsive design and UI components
- Axios: For handling HTTP requests to the backend API

Backend Framework:

- Flask: A lightweight Python web framework
- Flask-PyMongo: For MongoDB integration
- Flask-CORS: For handling Cross-Origin Resource Sharing
- JWT: For authentication and authorization

Database:

MongoDB: A NoSQL document database

Collections:

- donors: Stores donor information
- recipients: Stores recipient information
- donations: Tracks donation records
- admins: Manages administrator accounts

Development Tools :

- Git: For version control
- VS Code: As the primary code editor
- MongoDB Compass: For database visualization and management

Key Technical Features:

1. RESTful API Architecture:

- Clear separation between frontend and backend
- Stateless API endpoints for all CRUD operations
- JSON data format for all communications

2. MongoDB Integration:

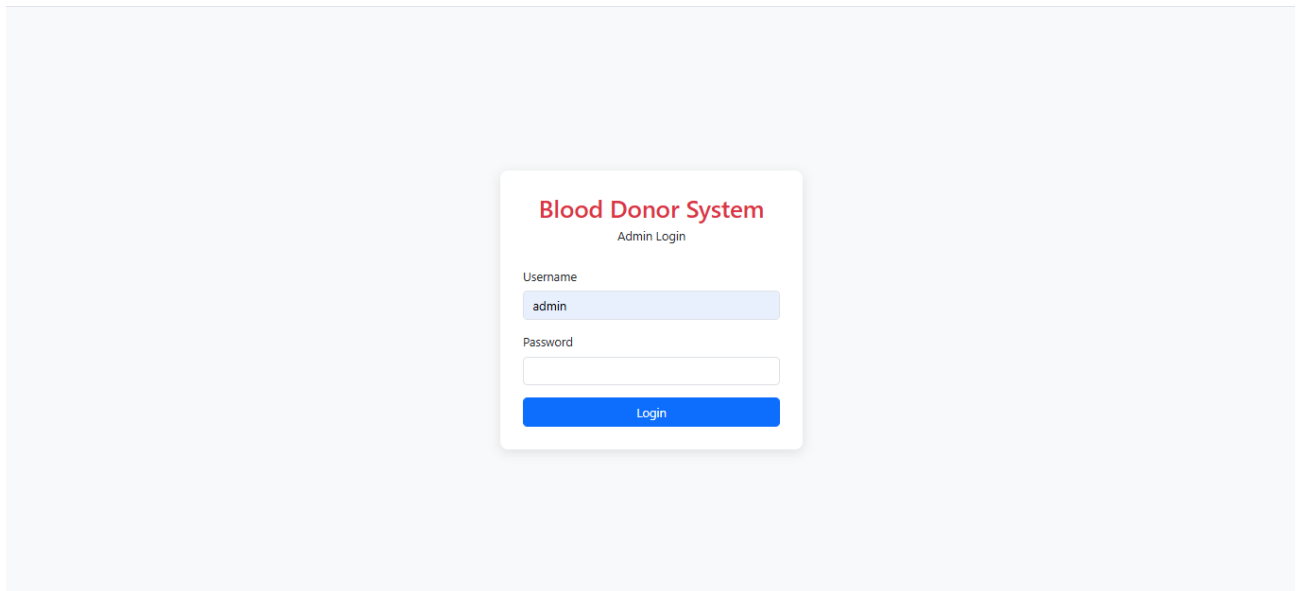
- Document-based data model
- Flexible schema design
- Embedded documents for related data
- Aggregation pipelines for analytics

3. Blood Matching Algorithm:

1. Based on standard blood type compatibility rules
2. Considers donor availability and location
3. Prioritizes matches based on urgency level

5. SCREENSHOTS

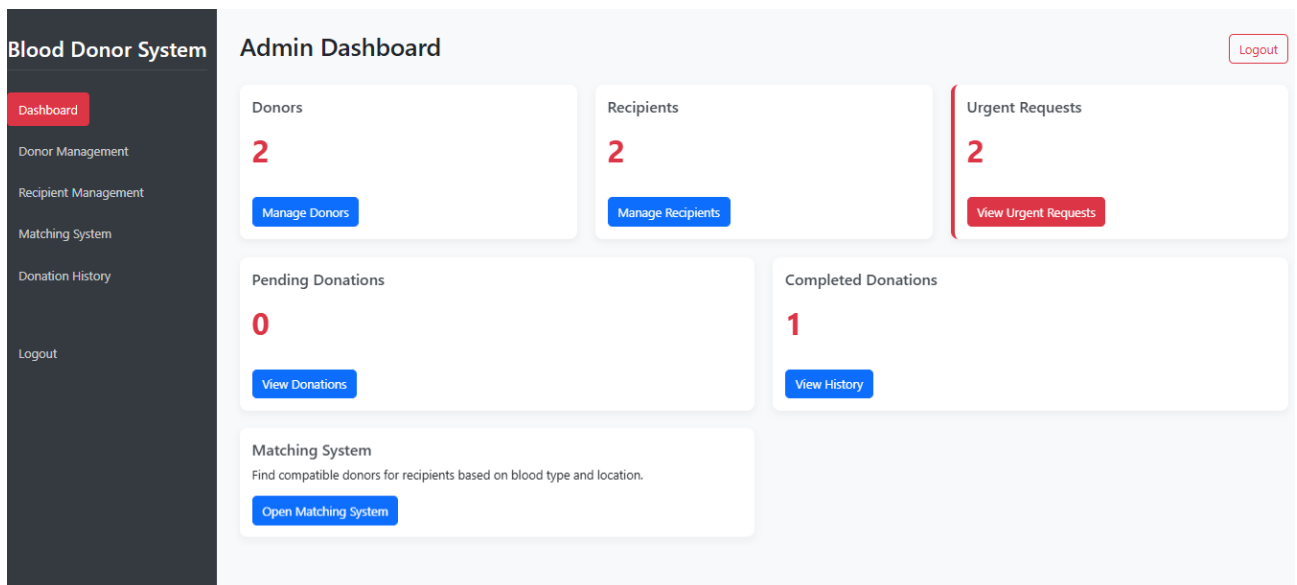
5.1 Login Page:



The screenshot shows the 'Blood Donor System' Admin Login interface. It features a central white card with a red title 'Blood Donor System' and subtitle 'Admin Login'. Below the title are two input fields: 'Username' with the value 'admin' and 'Password' which is empty. A blue 'Login' button is positioned at the bottom of the card. The background is a light gray gradient.

Figure 5. 1 : Admin login interface with secure authentication

5.2 Dashboard



The screenshot displays the 'Blood Donor System Admin Dashboard'. On the left is a dark sidebar with the system name and a list of navigation options: 'Dashboard' (highlighted), 'Donor Management', 'Recipient Management', 'Matching System', 'Donation History', and 'Logout'. The main content area is titled 'Admin Dashboard' and includes a 'Logout' button in the top right corner. It contains five data cards: 'Donors' (2, Manage Donors), 'Recipients' (2, Manage Recipients), 'Urgent Requests' (2, View Urgent Requests), 'Pending Donations' (0, View Donations), and 'Completed Donations' (1, View History). At the bottom is a 'Matching System' card with a description and an 'Open Matching System' button.

Figure 5. 2 : Main dashboard showing key metrics and navigation options

5.3 Donor Management

Blood Donor System

Dashboard

Donor Management

Recipient Management

Matching System

Donation History

Logout

Donor Management

Add New Donor

All Blood Groups

Search donors...

Name	Age	Blood Group	Location	Contact	Status	Actions
Joseph	20	A+	Banglore	1234567890 NO Field	Unavailable	Edit Delete
Usha	45	A+	Manglore	1234554321 NO Field	Available	Edit Delete

Figure 5. 3 : Donor listing with search and filter capabilities

5.4 Add/Edit Donor Form

Blood Donor System

Dashboard

Donor Management

Recipient Management

Matching System

Donation History

Logout

Donor Management

Edit Donor

Name

Usha

Age

45

Gender

Male

Blood Group

A+

Location/City

Manglore

Last Donated Date

05-10-2024

Email

NO Field

Phone Number

1234554321

Weight (kg)

Height (cm)

Occupation

NO Field

Available for donation

Cancel

Update Donor

Figure 5. 4 : Form for adding or editing donor information with validation

5.5 Recipient Management

Blood Donor System

Dashboard

Donor Management

Recipient Management

Matching System

Donation History

Logout

Recipient Management

Add New Recipient

All Blood GroupsAll Urgency LevelsSearch recipients...

Name	Blood Group	Hospital	Location	Date Needed	Urgency	Actions
Raju	A+	Sanjay Gandhi	Banglore	Invalid Date	High	EditDelete
Manu	A+	ST.Joseph	NO Field	4/19/2025	Medium	EditDelete

Figure 5. 5 : Recipient listing with urgency indicators

5.6 Add/Edit Recipient Form

Blood Donor System

Dashboard

Donor Management

Recipient Management

Matching System

Donation History

Logout

Recipient Management

Edit Recipient

Name

Manu

Blood Group Required

A+

Hospital

ST.Joseph

Location/City

NO Field

Urgency Level

Medium

Date Needed

19-04-2025

Contact Number

1234567892

Patient Age

Patient Gender

Male

Required Units

1

Doctor's Name

NO Field

Cancel

Update Recipient

Figure 5. 6 : Form for adding or editing recipient information

5.7 Matching System

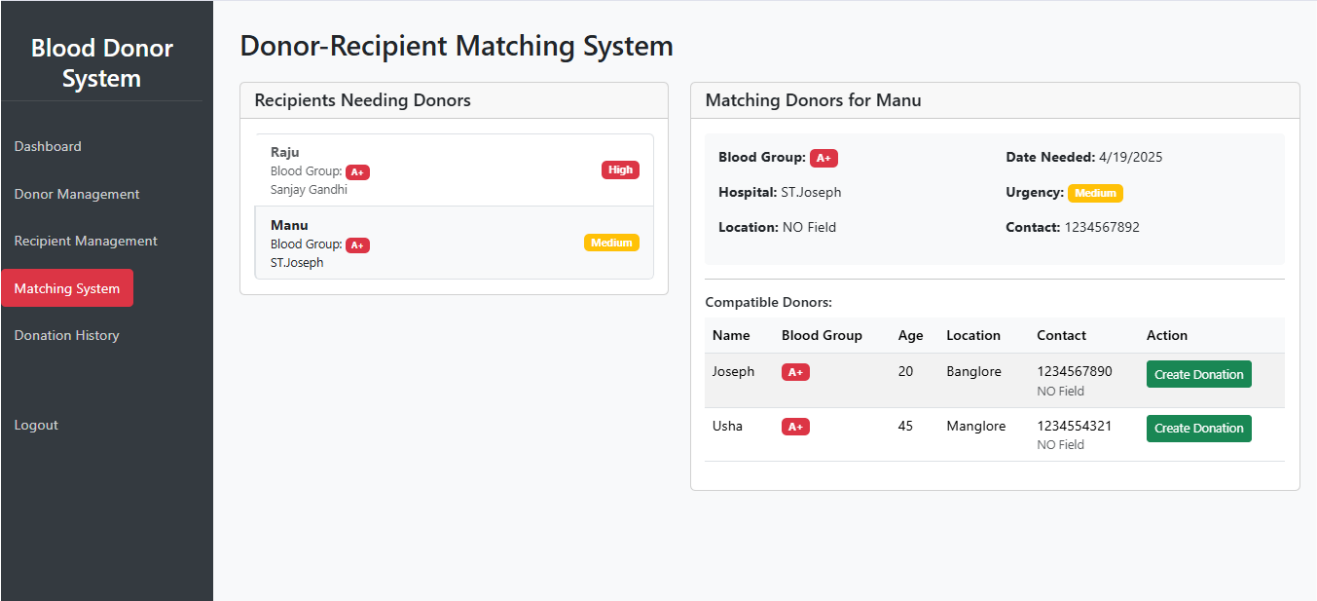


Figure 5. 7: Donor-recipient matching interface showing compatible donors

5.8 Donation History

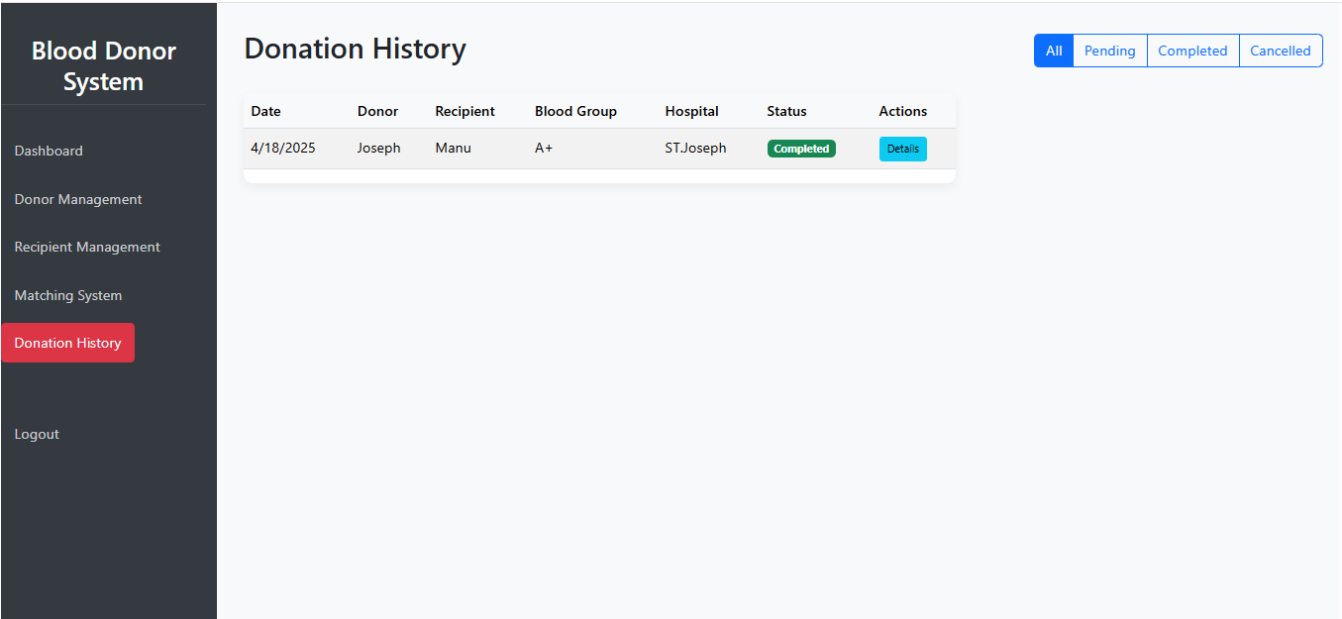


Figure 5. 8 : Donation tracking and history with status updates

5.9 MongoDB Collections

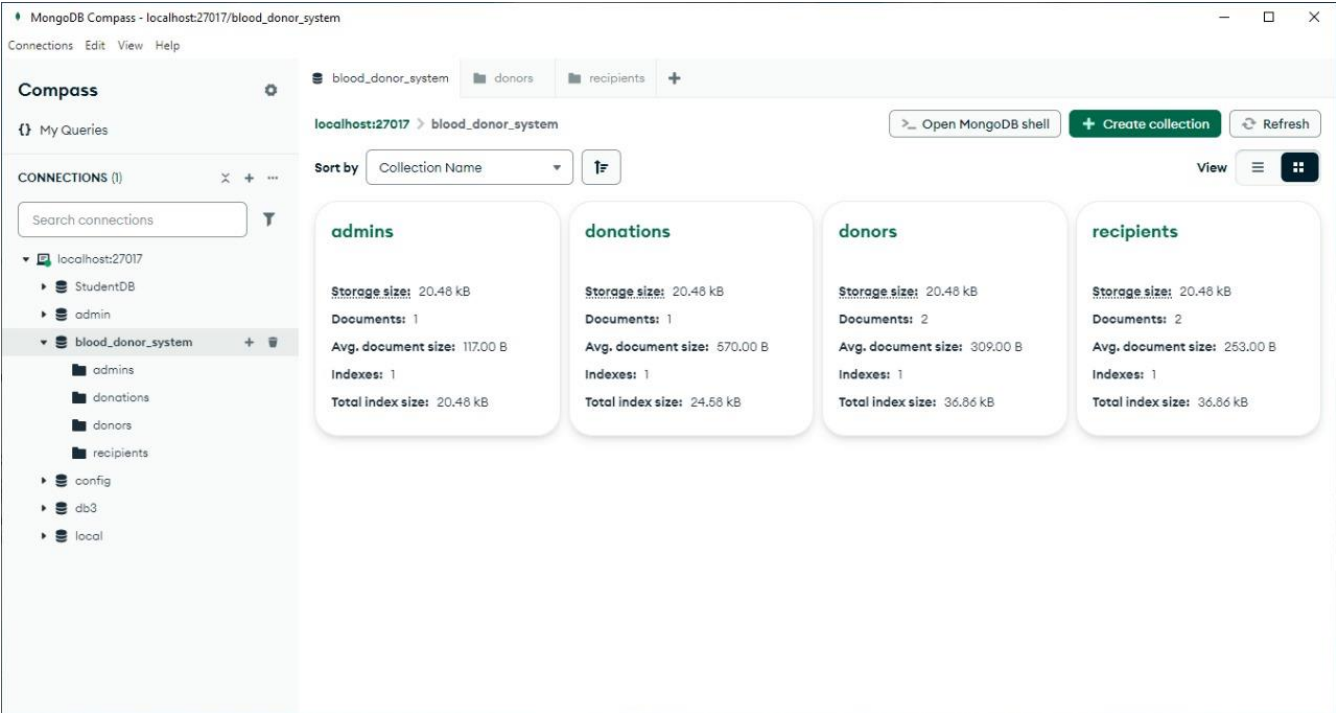


Figure 5. 9 : MongoDB Compass view showing the database collections

5.10 Donor Document Example

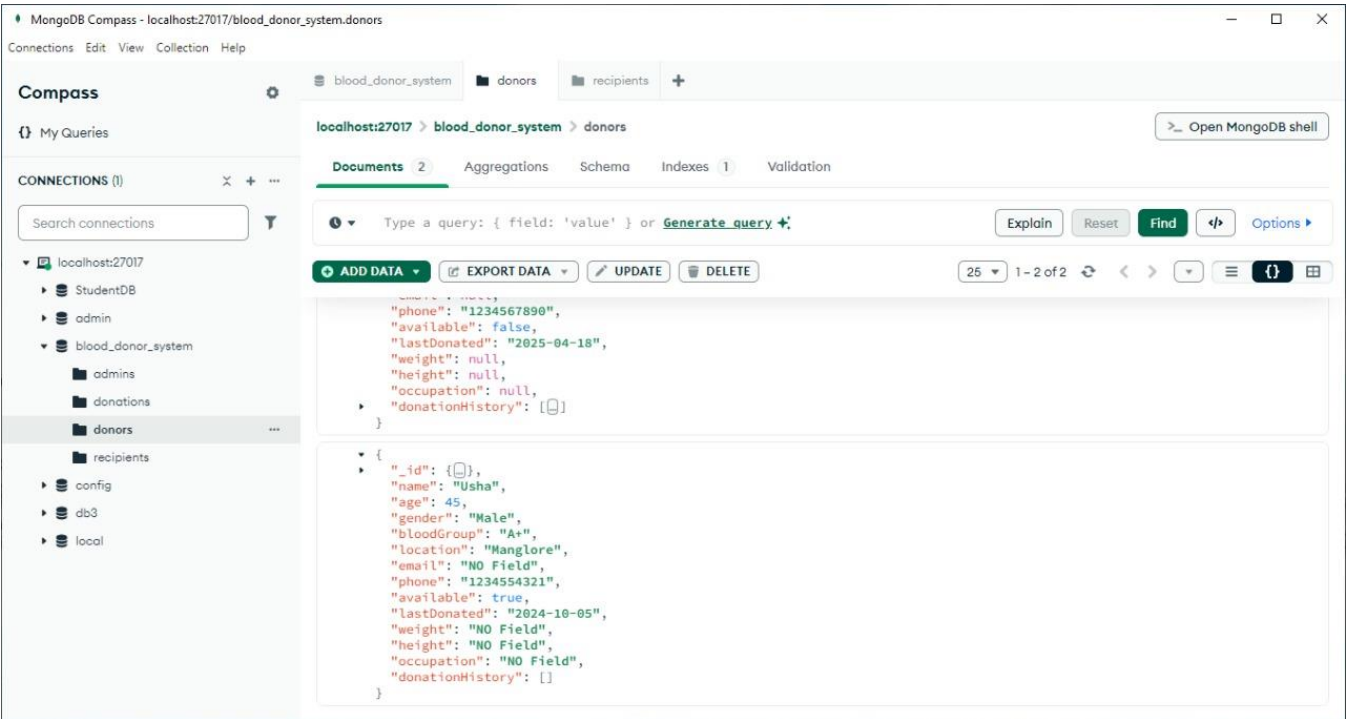


Figure 5. 9 : Example of a donor document in MongoDB showing the flexible schema

6. CONCLUSION

The Blood Donor Management System is a successful implementation of modern web technologies and NoSQL database design, addressing key healthcare challenges. It streamlines donor-recipient matching, enhances data accessibility, and supports informed decision-making for blood collection efforts. Despite challenges like schema validation, memory usage, and backup complexity in MongoDB, the system's performance, flexibility, and scalability have proven effective. With a user-friendly interface and centralized data management, it ensures quick adoption by healthcare staff and efficient donation tracking. Overall, the system offers a strong foundation for future enhancements and serves as a model for similar healthcare applications.

7. FUTURE SCOPE

The Blood Donor Management System has been designed with extensibility in mind, allowing for numerous enhancements and additional features in future iterations. The following areas represent promising directions for future development:

- 1. Mobile Application Development** : Introduce donor and staff apps with real-time alerts and appointment scheduling features.
- 2. Advanced Analytics and Reporting** : Use data to predict blood needs, track donation trends, and generate visual reports.
- 3. Integration Capabilities** : Enable seamless connection with hospitals, labs, and national blood bank systems.
- 4. Enhanced Security and Compliance** : Strengthen data protection and ensure adherence to healthcare regulations like HIPAA.
- 5. Artificial Intelligence and Machine Learning** : Leverage AI to improve donor-recipient matching and optimize donation schedules.